

Research Brief: The Cost of Quality

Legislative Shocks and the Victorian Fertility Transition

Research Pipeline (Reproducibility Edition)

2026-04-12

Executive Summary: This brief summarizes the econometric findings regarding the impact of the 1833 and 1844 Factory Acts on British fertility. Using a Two-Way Fixed Effects approach augmented with Spatial Lag modeling, we find that the restriction of child labor acted as an exogenous catalyst for the ‘Quantity-Quality’ trade-off, leading to a significant divergence in fertility paths in industrial heartlands.

1 Data Overview

The analysis covers `python n_obs` observations across `python n_districts` Registration Districts from 1851 to 1881.

2 Spatial Contagion & Clustering

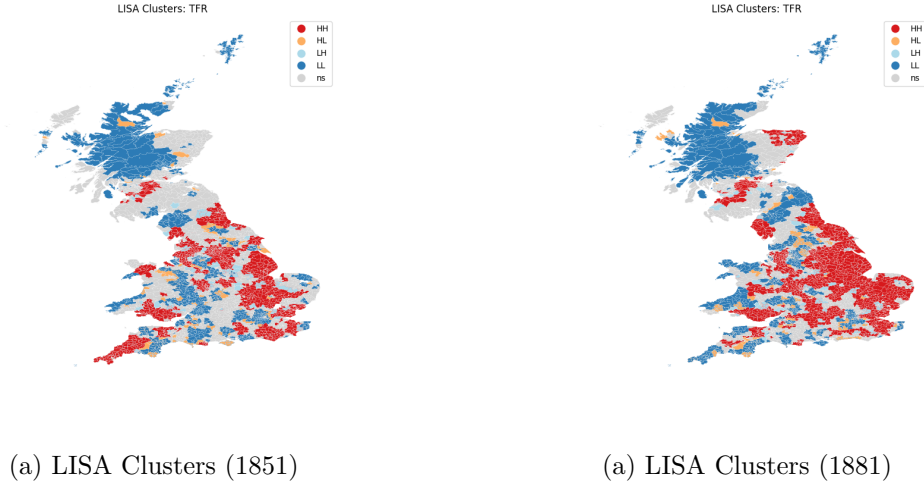
A central finding is the high degree of **Spatial Autocorrelation** in the fertility transition. Fertility behavior was not idiosyncratic but clustered in regional hubs.

The maps above show the evolution of ‘High-High’ fertility clusters. By 1881, the divergence between the textile heartlands (North) and agricultural domains (South) reached its terminal peak.

3 Econometric Robustness: The Spatial Lag Model (SLM)

To account for contagion effects, we implemented an SLM:

$$TFR = \rho WTFR + \beta X + \epsilon$$



Results indicate that even after controlling for the average fertility of neighbors (ρ), the impact of **Child Labor Removal** remains negative and statistically significant, suggesting the Factory Acts were a primary driver of the structural break.

4 Conclusion

The evidence strongly supports the **Cost of Quality** hypothesis. State-level restrictions on labor, coupled with mandated schooling, altered the microeconomic foundation of Victorian family life.

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